Rapid Prototyping Report

Trigonometric Wheel - Report

Group H

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1. Introduction

The Trigonometric Wheel is an innovative educational tool designed to simplify mathematical calculations involving angles, and trigonometric functions. Targeted at students and educators, the wheel aims to combine functionality, affordability, and aesthetic appeal. Branded with the TalTech logo, it is positioned as a high-value souvenir for the university's shop, serving both educational and promotional purposes.

2. Final Idea of TalTech Souvenir

The Trigonometric Wheel features three interlocking circular layers, enabling users to perform calculations involving trigonometric values. It incorporates clear angle measurements and displays precise values of cosine, sine, tangent, and cotangent for commonly used angles. This product aligns with TalTech's values of innovation, precision, and educational excellence.

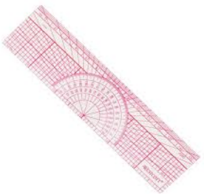
2.1. Design Language

The design language of this product embodies a thoughtful and systematic approach, driven by clearly defined principles and characteristics to ensure its functionality, relevance, and appeal. The design drivers focus on delivering soft values that resonate with users, emphasizing informativeness and utility, simplicity paired with a holistic perspective, and a commitment to fostering growth and innovation. A cornerstone of this approach is precision and accuracy, reinforcing the product's role in enabling future professionals to achieve tangible success in their endeavors.

The desired attitude of the product is encapsulated in a set of core attributes: it is precise, reliable, and logical, while remaining straightforward and honest in its presentation. These qualities are complemented by a compact and stylish aesthetic, ensuring both practicality and visual appeal. To guide the visual identity, a curated style board references elements from unrelated industries or product categories, fostering creative inspiration and ensuring a unique, cross-disciplinary visual appeal. This strategy promotes innovation while adhering to the product's intended functional and emotional impact.

Visual style board:









2.2. Trigonometric Wheel Technical Specifications

The Trigonometric Wheel is an educational tool designed to make it easier for students and teachers to calculate angles and radians. This device seeks to combine essential mathematical functionalities with an intuitive, practical and accessible design, becoming a useful resource for learning in areas of engineering and mathematics.

In its design, the Trigonometric Wheel will include several features intended to maximize its usefulness:

- It will offer trigonometric values such as sine, cosine, tangent and cotangent for commonly used angles, such as 0°, 30°, 45°, 60°, 90°, 180°, 270°.
- It will incorporate clear and precise angular measurements that facilitate calculations.
- All this will be accompanied by an attractive visual design that will include the TalTech logo, also guaranteeing an affordable price for users.

Project Objectives

The main objective of the Trigonometric Wheel is to become an effective and accessible learning resource. To achieve this, the project has set concrete goals. Firstly, it seeks to provide in a clear and visually appealing way the trigonometric values mentioned. Likewise, a balance between functionality, aesthetics and cost is sought so that this tool is viable in both educational and commercial environments.

In terms of feasibility, the development team will work with experts in design and education, ensuring that the final product meets the necessary pedagogical standards and is produced within the planned budget. Finally, the Trigonometric Wheel must be fully designed and ready for production no later than December 21, 2024.

Product Requirements

The design of the Trigonometric Wheel must meet the following criteria:

- Include trigonometric values of key angles, providing the user with a quick and reliable reference.
- Incorporate precise angular markings that are easy to interpret.
- Present an aesthetically appealing design that reflects TalTech's identity by including its logo.
- Ensure that its production cost allows for an affordable selling price.

Excluded Areas

The development of the Trigonometrical Wheel will focus solely on basic mathematical functions, such as elementary trigonometry. Advanced functions such as logarithms or

exponential calculations will not be contemplated. Likewise, digital versions, mobile applications, or specific adaptations for different educational levels will not be included.

Impact Assessment

The impact of the Trigonometric Wheel will be measured through various indicators. Surveys and feedback from students and teachers will be collected to assess its usefulness in learning. In addition, improvements in the understanding of mathematical concepts related to trigonometry will be analyzed. Finally, sales and customer feedback will serve as a reference to measure the acceptance of the product in the market.

In conclusion, the Trigonometric Wheel represents an effort to offer an innovative, functional and accessible educational tool that responds to the needs of users and enhances the learning experience in academic environments.

2.3. Electronic Integration

Integration of an LCD Display for Dynamic Calculations

- Function Improvement: An LCD screen could be added to display calculations dynamically. Instead of static numbers, the wheel could include input buttons or a touch interface allowing users to select numbers, angles, and operations (e.g., cosine, sine), and the device would automatically calculate and show results on the screen.

- Technical Complexity: This would introduce complexity, as it requires the integration of a microcontroller, display, and input buttons, along with basic programming for trigonometric functions.

- Influence on Price: The cost of the product would increase due to the need for electronics (LCD screen, microcontroller, buttons) and a power source, making it pricier but more advanced.

Voice Command Functionality

- Function Improvement: By incorporating a small speaker and voice recognition system, the wheel could allow users to verbally ask for specific calculations (e.g., "What is the cosine of 45 degrees?"). The electronic device would then respond with the correct result through the speaker.

- Technical Complexity: Adding voice recognition requires more sophisticated electronics, including a microphone, speaker, and software for processing voice commands. This increases the technical challenge significantly.

- Influence on Price: The price would rise further due to the additional hardware and development costs for speech recognition, positioning the product as more premium and technology-driven.

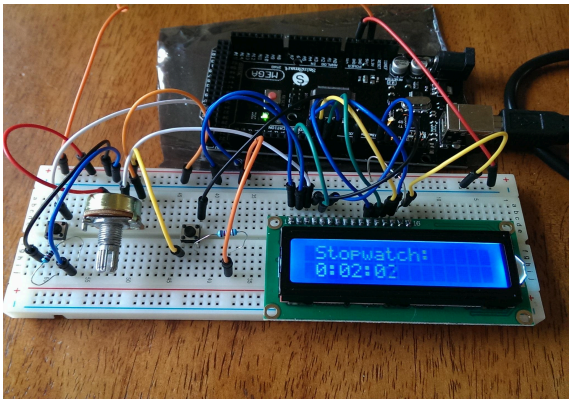
LED Lights for Visual Representation

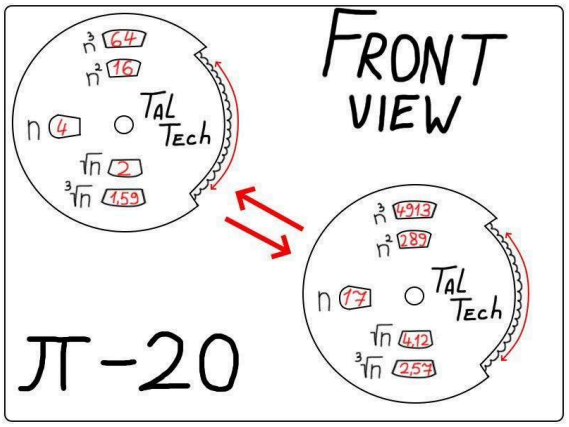
- Function Improvement: Integrating LED lights around the wheel to visually guide users to the correct answer or highlight certain values could make learning more interactive. For example, when calculating a trigonometric value, the corresponding section of the wheel could light up to show the result.

- Technical Complexity: This would require a basic circuit with LEDs, perhaps controlled by a small microcontroller. Compared to the previous ideas, this is a simpler and more cost-effective addition.

- Influence on Price: The price increase would be moderate since LEDs are inexpensive, though it would still require additional design and assembly work to incorporate electronics.

Each of these additions would improve the educational value of the product, making it more interactive and appealing to tech-savvy users, but the trade-offs would involve increased complexity and a higher price.





In Photos to demonstrate, we have access to arduino boards (left) to create a prototype of electrical components but to get them to the size of the scale of our design (right) which is about 15 cm in diameter, it is not possible with the resources we know of at our school.

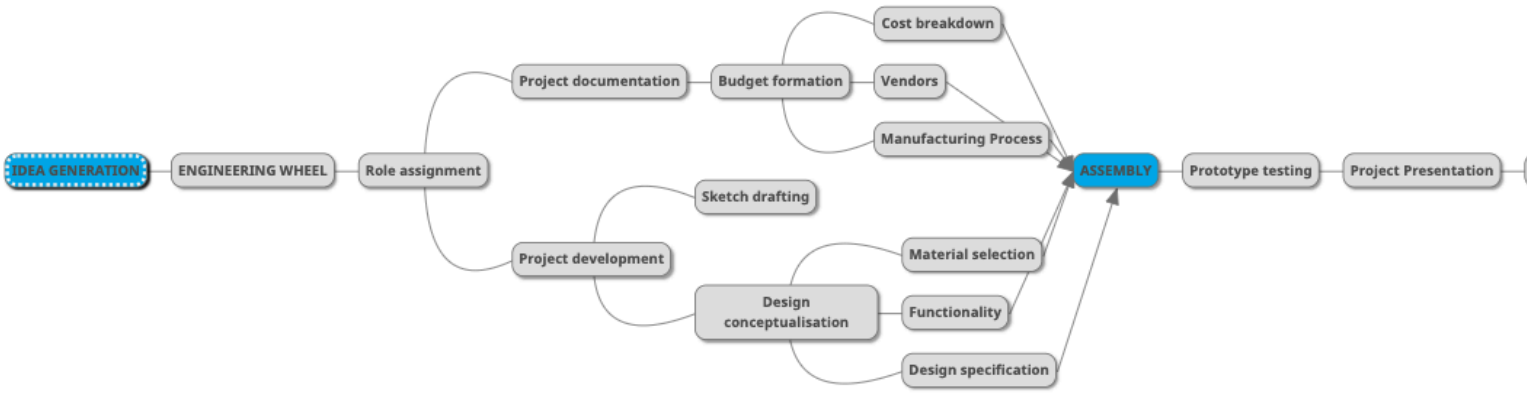
Conclusion

We do not have the financial or time resources to implement an electric component to our project. Adding electronic parts would make producing the prototype difficult for us to get to the scale we like with our limited resources (see photos above of example). However, it would make the product more consumer-friendly and precise.

3. Project Management

3.1. Work Breakdown Structure (WBS)

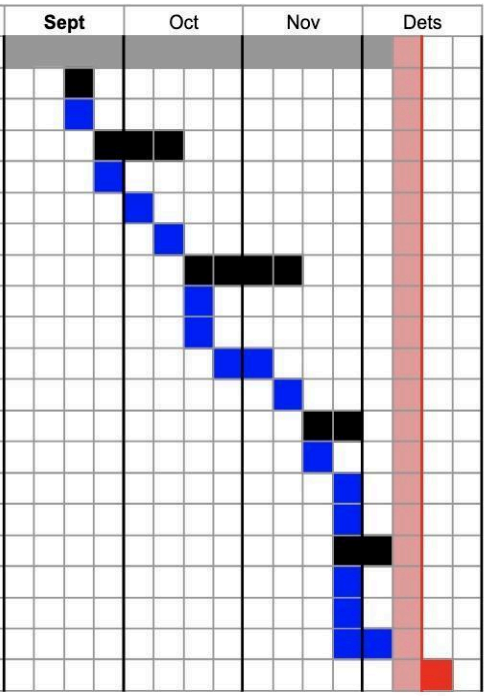
1. Research
    - Literature review on existing tools.
    - User needs analysis.
2. Prototype Development
    - Functionality and design.
    - Testing and iteration.
3. Study and Analysis
    - Data collection and instrument development.
    - Detailed analysis and reporting.
4. Manufacturing
    - Optimizing design for production.
    - Identifying manufacturers.
    - Organizing logistics.
5. Marketing and Public Relations
    - Creating promotional materials.
    - Launching a marketing campaign.



3.2. Gantt Chart

A detailed Gantt chart outlines key milestones, starting with research (September 16, 2024) and concluding with the marketing campaign (December 7, 2024). It ensures a structured approach to project execution.

Task	Dur W	Start	End	Work H	Resources
Wheel with powers/roots/angles	**16**	**16.09.24**	**07.12.24**	76	
1 Research	**1**	**16.09.24**	**21.09.24**	10	
1.1 Research	1	16.09.24	21.09.24	10	All
2 Prototype Development	**3**	**22.09.24**	**12.10.24**	16	
2.1 Functionality	1	22.09.24	28.09.24	3	Developer
2.2 Design	1	29.09.24	05.10.24	10	Developer/Lead
2.3 Testing	1	05.10.24	12.10.24	3	Developer
3 Study/analysis	**5**	**13.10.24**	**09.11.24**	19	
3.1 Data collection and Analysis plan	1	13.10.24	19.10.24	5	Analyst
3.2 Data Collection Instruments	1	13.10.24	19.10.24	2	Analyst
3.3 Data Collection	2	20.10.24	02.11.24	7	Analyst
3.4 Analysis Report	1	03.11.24	09.11.24	5	Analyst
4 Manufacturing	**3**	**10.11.24**	**23.11.24**	19	
4.1 Optimizing design	1	10.11.24	16.11.24	4	Developer
4.2 Finding Manufacturer	1	17.11.24	23.11.24	10	Manager/Financial
4.3 Logistics	1	17.11.24	23.11.24	5	Manager
5.Public Relations	**4**	**24.11.24**	**07.12.24**	12	
5.1 Project development	1	24.11.24	30.11.24	2	Developer
5.2 Flyer Production	1	24.11.24	30.11.24	4	Artist
5.3 Marketing	2	24.11.24	07.12.24	6	All
End	**0**	**07.12.24**	**07.12.24**	**0**	

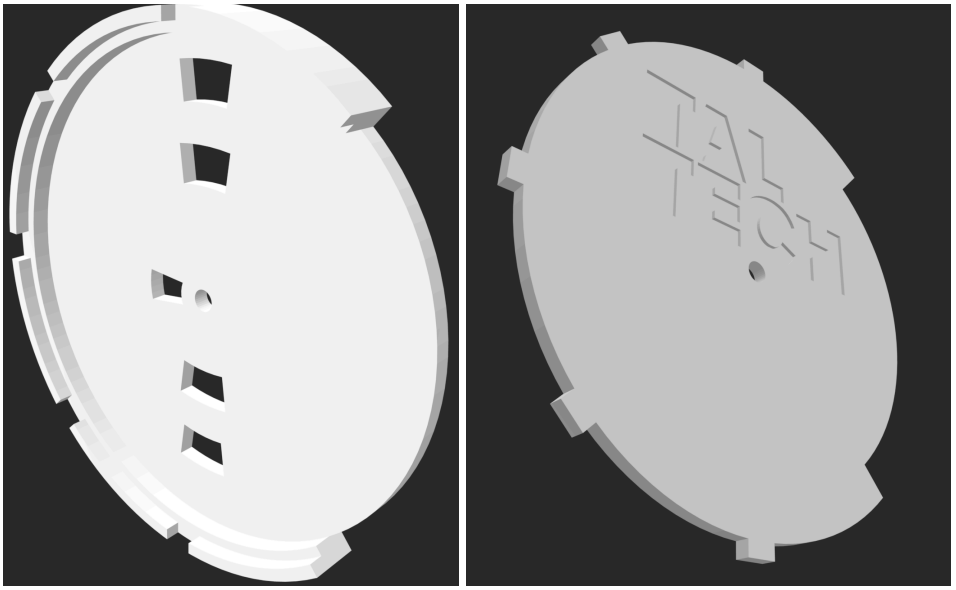


4. CAD Models of the Product

Detailed CAD models were developed using 3D design software to illustrate:

1. The interlocking mechanism.
2. Layer configurations.
3. Placement of trigonometric and numeric data.

These models prioritize ease of use, manufacturability, and a clear display of mathematical values, ensuring the product's effectiveness as an educational tool.



5. Suitable Materials

Material Selection:

- **Primary Material:** PLA filament for 3D printing, chosen for its affordability, durability, and ease of use.
- **Decal Material:** Sticker paper for applying the TalTech logo and markings.
- **Adhesive:** High-quality glue for securing decals.

Rationale:

- Cost-effective for batch production.
- Environmentally friendly.
- Ensures durability and aesthetic appeal.

6. Production Technologies

Prototyping

- Utilizes 3D printing with PLA filament for precision and rapid iteration.

Batch Production

- CNC machining for large-scale manufacturing.
- Application of decals using automated processes for consistency.
- Consideration for injection molding as demand scales up.

7. Business Model Including Economic Aspects

7.1. Financial Plan

Research on Materials and Costs

1. PLA Filament:
    - PLA is the material used for 3D printing.
    - Density: 1.24 g/cm³.
    - Volume of 3 Wheels:
        - Diameter: 7.5 cm → Radius: 3.75 cm.
        - Area per wheel: π × (3.75²) ≈ 44.18 cm².
        - Thickness of one wheel: 1.2 cm ÷ 3 = 0.4 cm.
        - Volume of one wheel: 44.18 × 0.4 ≈ 17.67 cm³.
        - Total volume (3 wheels): 17.67 × 3 ≈ 53.01 cm³.
    - Mass: Volume × Density = 53.01 × 1.24 ≈ 65.73 g.
    - Cost:
        - PLA filament: €18/kg.
        - Cost per wheel: €18 × (65.73 ÷ 1000) ≈ €1.18 per wheel.
2. Sticker Paper for Decals:
    - Sticker paper will be used to print logos and markings.
    - Cost:
        - €9 for 20 A4 sheets.
        - One sheet can accommodate decals for 10 wheels.
        - Cost per wheel: €9 ÷ 20 × (1 ÷ 10) ≈ €0.45 per wheel.
3. Glue for Decals:
    - Glue will attach the decals to the wheel.
    - Cost:
        - €4.50 per bottle, sufficient for 100 wheels.
        - Cost per wheel: €4.50 ÷ 100 = €0.045 per wheel.

Fixed and Variable Costs

1. Fixed Costs:
    - No Fixed Costs:
        - Since the school provides the printers and maintenance free of charge, there are no fixed costs for this project.
2. Variable Costs per Unit:
    - PLA Filament: €1.18.
    - Sticker Paper: €0.45.
    - Glue: €0.045.

    - Total Variable Cost per Wheel: €1.68.
3. Cost Price of the Product:
    - Since there are no fixed costs:
        - Total Cost Price per Wheel: €1.68 per wheel.

Break-even Point and Production Plan

1. Selling Price:
    - Each wheel will sell for €10.
2. Profit per Wheel:
    - Selling price - Total cost price = €10 - €1.68 = €8.32 per wheel.
3. Wheels Required for Monthly Salaries:
    - Each team member earns €2000 per month.
    - Total salary for 5 members = €2000 × 5 = €10,000 per month.
    - Wheels needed = Total salary ÷ Profit per wheel.
    - €10,000 ÷ €8.32 ≈ 1,202 wheels per month.
4. Daily Production Goal:
    - Monthly production: 1,202 wheels.
    - Daily production (assuming 30 days/month): 40 wheels per day.
5. Annual Production Goal:
    - 1,202 wheels × 12 months = 14,424 wheels per year.

Summary

- Selling Price: €10 per wheel.
- Variable Cost per Wheel: €1.68.
- Fixed Costs: None.
- Profit per Wheel: €8.32.
- Wheels Required for Monthly Salaries: 1,202 wheels per month.
- Daily Printing Goal: 40 wheels per day.
- Annual Production Goal: 14,424 wheels per year.

7.2. Business Model Canvas

Key Partners

- **TalTech School Administration**: Support for 3D printing and resources.
- **Teacher Mentors**: Guidance on design, production, and marketing.
- **Gift Store (On-campus and Online)**: Sales and distribution channel.
- **Material Suppliers**: Source for filaments and other 3D printing materials.
- **Student Community**: Feedback and promotion among peers.

Key Activities

- **Design**: Use CAD software to create the Engineering Wheel.
- **Production**: Manufacture using school's 3D printers.
- **Quality Assurance**: Test functionality, durability, and accuracy.
- **Marketing**: Promote via social media, posters, and school events.
- **Sales**: Manage inventory and handle sales in the gift store and online.

Key Resources

- **Human Resources**: Team members with CAD and engineering knowledge.
- **Equipment**: School's 3D printers and computers with CAD software.
- **Mentorship**: Support from teachers and experts in engineering and business.
- **Materials**: Filaments and consumables for 3D printing.

Value Propositions

- Affordable, high-quality tool for engineering calculations.
- Simplifies tasks like angle measurement, trigonometric values, and power calculations.
- Visually appealing and branded with the TalTech logo.
- Eco-friendly production using local 3D printers and minimal waste.
- A unique product that showcases students' skills and supports education.

Customer Relationships

- **Student-focused Approach**: Engage through clubs, workshops, and fairs.
- **Feedback Loop**: Use surveys and reviews to improve the product.
- **Community Engagement**: Promote it as a project supporting peer education.

Channels

- **On-Campus Gift Store**: Physical presence for students and visitors.
- **Online Store**: Access for alumni, external customers, and wider audiences.
- **Social Media and University Platforms**: Promotion and communication.
- **Events**: Showcasing at fairs, open days, and student exhibitions.

Customer Segments

- **Primary**: TalTech students (engineering, physics, and math disciplines).
- **Secondary**: Faculty and researchers needing a quick reference tool.
- **Tertiary**: High school students and alumni interested in TalTech-branded items

Cost Structure

- **Fixed Costs**: 3D printer maintenance, filament stock, software licenses (if any).
- **Variable Costs**: Materials per unit, electricity for printing.
- **Marketing Costs**: Posters, online ads, or event participation fees.
- **Distribution Costs**: Packaging for online orders.

Revenue Streams

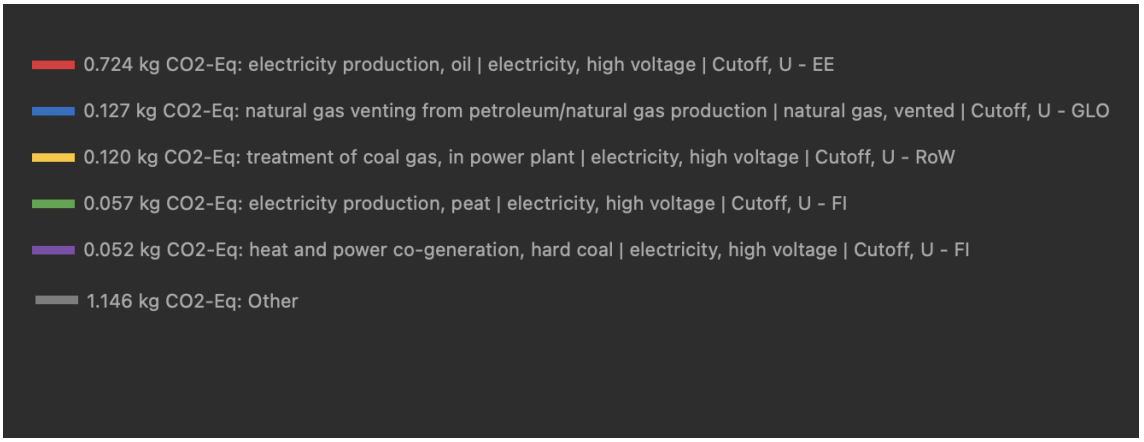
- **Direct Sales**: From gift store and online store purchases.
- **Custom Orders**: Special designs or bulk orders for classes or groups.
- **Sponsorships**: Partnerships with alumni or companies for additional funding.

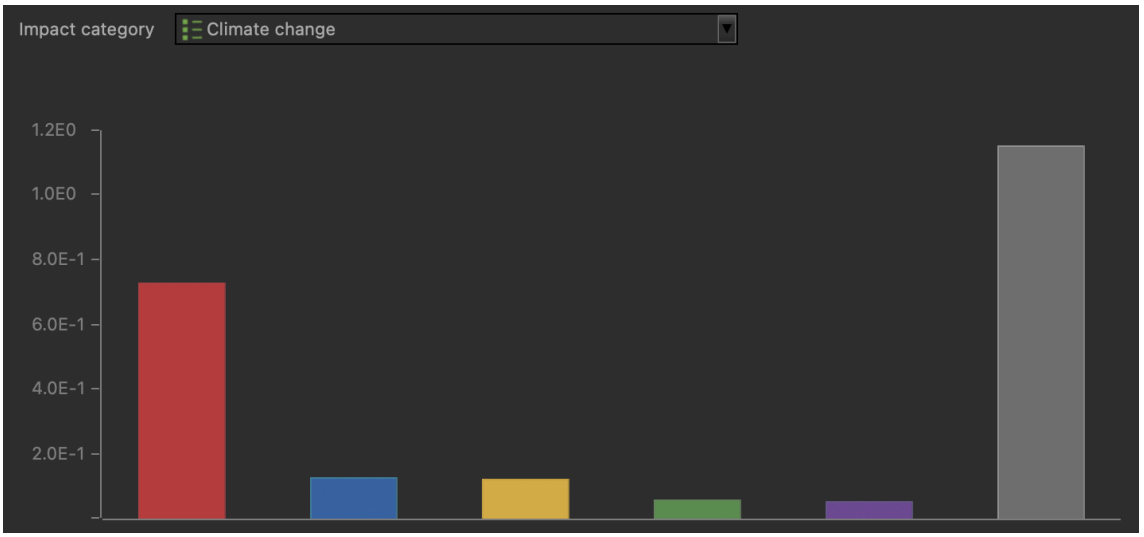
8. Overview and Results of Life Cycle Assessment (LCA)

Key Findings:

1. **Material Impact:**
    - PLA filament has a relatively low environmental footprint.
2. **Manufacturing Energy Use:**
    - 3D printing energy consumption was assessed and optimized.
3. **End-of-Life:**
    - PLA's biodegradable properties ensure minimal waste impact.

Conclusion: The Trigonometric Wheel is designed to align with sustainable practices, minimizing its ecological footprint. Regular assessments ensure continuous improvement in reducing environmental impacts.





9. Individual Contributions

Resource Plan:

- **Artur Alen (Project Lead):** Oversees project execution.
- **Alejandro García Álvarez (Project Manager):** Ensures timelines are met and coordinates activities.
- **Maksim Sauliak (Developer):** Focuses on prototype development and design refinement.
- **Jordan Brodie (Financial Advisor):** Manages budget and cost analysis.
- **Arthur Andreas Aavik (Market Analyst):** Conducts market research and analysis.

Alejandro García Álvarez:

- Led project planning and timeline management.
- Coordinated team activities and ensured milestones were met.

Artur Alen:

- Directed overall project vision and ensured alignment with objectives.
- Assisted in refining the prototype and overseeing functionality tests.

Maksim Sauliak:

- Developed and tested prototypes.
- Refined design based on user feedback.

Jordan Brodie:

- Conducted financial analysis and ensured budget adherence.
- Evaluated cost efficiency for batch production.

Arthur Andreas Aavik:

- Researched market potential and contributed to marketing strategy.
- Gathered data for LCA and provided analytical insights.

The team's success is attributed to the collective efforts of each member, with distinct roles and responsibilities. We all shared responsibilities and contributed to the project. We didn't stick just to our assigned roles oftentimes we crossed over.

.	Project lead	Project manager	Financial advisor	Developer	Market Analyst
Needs	**R**	**S**			
Functions	**R**	**S**		**S**	
Plan	**R**	**S**	**S**	**S**	**S**
Design	**S**	**R**		**S**	
Development	**S**	**S**		**R**	
Finances	**S**		**R**		**S**
Test	**S**	**R**		**S**	
Analyst	**S**		**S**		**R**

10. Conclusion

The Trigonometric Wheel embodies TalTech's commitment to innovation, education, and sustainability. It serves as a practical tool for students and a memorable souvenir, combining functionality with a strong brand identity. With a clear production and marketing strategy, it is positioned for success in the university's souvenir shop and beyond.